SCI-NOI v0.1

SCI-NOI v0.1 Stage B Scientific Rationale

Implementation-blind Stage B draft

SCI-NOI_SCIENTIFIC_RATIONALE v0.1/draft-r0.1

Not owner-accepted or frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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1. Why the three NOI roles must remain separate

SCI-NOI describes three different questions.

- NOI-GEN asks what finite collection of conditional randomizations was generated from one exact immutable parent and one exact operator graph.
- NOI-UNC asks what empirical quantity a named estimator infers from one admitted ensemble.
- NOI-STD asks how one immutable signal numerator is divided by one compatible positive uncertainty scale.

A realization ensemble is not an uncertainty estimate merely because it is called a noise or jackknife ensemble. An empirical second moment is not a covariance merely because it is stored as a pointwise field. A reciprocal second-moment scale is not inverse variance or precision. A dimensionless standardized map is not a Gaussian significance map. These distinctions are the central scientific safety boundary of this contract.

2. The ordinary conditional-randomization question

The ordinary route is NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1. It begins with exact retained PTC occurrences and inserts an NOI-owned sign modifier at the exact PTC-to-MAP numerical boundary. The complete frozen MAP accumulation then produces an NOI realization map. MAP may apply the modifier inline, or the same operation may be materialized, but the scientific object is the same only when the identical modifier acts on the identical admitted occurrence before identical frozen MAP arithmetic. The output is not an ordinary MAP science product.

For member b, the fixed-state relation is

```
M_b = O_Theta0(R_b(parent)).
```

The learned state Theta0 is fixed. A procedure that instead derives Theta_b from each randomized parent asks another scientific question:

```
Theta_b = LearnResolve_b(R_b(parent)),
M_b     = O_Theta_b(R_b(parent)).
```

Those members must have another method identity and cannot enter the same uncertainty estimate as fixed-state members.

3. Coherence and network-local balance

The ordinary coherence unit is one stable realized detector/channel within one exact observation. One assigned sign applies to every admitted PTC occurrence for that detector in that observation. Time, scan, chunk, traversal, worker, and container boundaries cannot change the assignment.

For each observation/readout-network stratum h, let D_h be the canonical ordered detector population. The exact frozen MAP-admitted positive contributions define the detector coefficient mass

```
B_d = sum_p sum_{i in C_p, detector(i)=d} G_pi gamma_i,   B_d > 0.
```

For a candidate sign vector s_h, define the dimensionless imbalance

```
Delta_h(s_h) = abs(sum_{d in D_h} s_d B_d) / sum_{d in D_h} B_d.
```

The Stage B design admits s_h exactly when

```
abs(sum_d s_d B_d) <= tau_h sum_d B_d,
```

where tau_h is an explicitly requested exact rational in [0,1). There is no inferred value. The comparison is performed on the exact rational values of the persisted numerical parent coefficients, not on traversal-dependent floating-point reductions.

The target law is uniform over the admissible sign vectors in each stratum, with independent product composition across strata and observations. The admissible set is complement-closed because Delta_h(s_h)=Delta_h(-s_h). Therefore every detector has marginal sign probability 1/2, while detector signs within a balanced stratum are generally dependent. Equal positive and negative detector counts are neither required nor implied.

This conditional law is realized by plan-bound rejection sampling from uniform independent sign candidates. The exact random-bit generator, algorithm version, key namespace, seed/key bytes, requested member count, and positive retry cap are required plan facts. The contract supplies no default for any of them. Candidate rejections are construction outcomes, not members. Failure to find an admissible candidate within the cap fails design resolution closed.

Members are sampled with replacement. Complement pairs are not forced. Duplicates and complements remain distinct draws and are retained; exact assignment equality, complement-orbit equality, counts, and design rank are reported separately. Equal design weights omega_b=1/B_resolved are part of this exact design, not inferred from a filename or count.

4. What the ensemble contains

The scientifically readable term selected here is source-bearing conditional randomization ensemble.

"Source-bearing" means that the fixed parent may contain astronomical source signal, structured residuals, and source-model error. "Conditional" means the declared parent and operator state are held fixed. "Randomization" identifies the assignment law, not repeated physical noise. The design intends to suppress source signal in aggregate, but it does not by construction establish source-free maps or pixelwise cancellation.

Writing a fixed parent as x=s+n for reasoning, the conditional randomization second moment contains terms from s s^T, n n^T, and their cross terms after the sign law and frozen operator act. A compact source can therefore leave a localized second-moment imprint. A structured scan residual can leave a coherence-dependent imprint. Removing such morphology is not sufficient to prove physical-noise fidelity because a learned source model may also absorb genuine noise modes.

5. The initial UNC estimand

Let the all-members-successful ensemble contain admitted realization maps M_b. On the exact common domain where every admitted member provides a valid finite value, the initial estimator is

```
V_hat_cond(p) = sum_b omega_b M_b(p)^2,
sum_b omega_b = 1,
omega_b = 1 / B_resolved.
```

The center is the known design target zero. The finite ensemble mean is not subtracted, and no B-1 correction applies. Dependence, complement structure, all distinct counts, design rank, use-specific effective information, and the uncertainty of V_hat_cond or its unavailable state remain explicit.

V_hat_cond is a conditional randomization second moment in squared signal units. It retains source imprint and structured residual content. Pointwise shape does not make it physical-noise variance or MAP covariance. Unreported covariance is unknown or unavailable, not zero or independence.

The exact reciprocal product is

```
W_hat_cond(p) = 1 / V_hat_cond(p)
```

only where the parent is finite and strictly positive. It is an inverse conditional second-moment scale in inverse squared signal units, not inverse variance or precision. No floor, cap, epsilon, clipping, or shrinkage is implicit.

6. The initial STD product

The initial standardized product binds the exact immutable normalized real-observation MAP signal q_MAP associated with the same frozen operator state:

```
sigma_cond(p) = sqrt(V_hat_cond(p)),
S_cond(p)     = q_MAP(p) / sigma_cond(p).
```

It exists only on the exact compatible intersection where numerator and scale are valid and the scale is finite and strictly positive. Its unit is exactly 1. Its complete claim is only: "MAP signal standardized by the stated conditional randomization second-moment scale." Numerator/scale dependence is part of the identity. No Gaussian, Student, z, N-sigma, probability, detection, completeness, purity, or catalog interpretation follows.

7. Immutable successors and externally owned transforms

An externally owned deterministic transformation may be applied to every compatible realization only under an exact content-bound authority and parity interface supplied by the process that owns the transformed science product. NOI does not choose, tune, simplify, relocate, or reinterpret it.

A Wiener transform frozen before realization application is one fixed owner-transformed route. A Wiener transform learned or updated from UNC_k begins a new immutable transformation, science-product, GEN, and UNC generation. Per-member learning is a distinct relearned method.

The same discipline applies to FRUIT. Fixed residual or terminal state supports only uncertainty conditional on that frozen FRUIT state. NOI-informed continuation creates a successor generation, and per-member replay is a separate relearned method. Prior NOI products remain dependent inputs rather than independent validation.

8. Analytic predictions of the draft contract

These are mathematical predictions of the stated design and estimators. They are not reports of implementation conformity or empirical validation.

- NOI-PRED-001: Globally complementing an admitted sign assignment preserves every stratum imbalance and produces an equally admissible assignment.
- NOI-PRED-002: Under the declared uniform complement-closed law, every detector has marginal sign probability 1/2 even though network-local signs are dependent.
- NOI-PRED-003: A one-detector stratum has Delta_h=1; it is infeasible for every permitted tau_h<1 and therefore fails design resolution closed.
- NOI-PRED-004: Two equal-mass detectors with opposite signs have Delta_h=0 and satisfy every permitted nonnegative tolerance.
- NOI-PRED-005: A candidate rejected before admission changes no requested,

resolved, completed, unique, or UNC-admitted member count.

- NOI-PRED-006: Exact duplicate assignments reduce B_unique but remain distinct with-replacement draws and retain their declared equal weights.
- NOI-PRED-007: Complement assignments are distinct full-distribution draws but produce the same member outer product under a fixed linear operator.
- NOI-PRED-008: Any admitted-member failure makes the whole GEN ensemble unavailable for UNC; completed survivors produce no partial estimate.
- NOI-PRED-009: V_hat_cond(p) is nonnegative wherever it is available and is unavailable outside the common all-member domain.
- NOI-PRED-010: A positive source-bearing fixed component may contribute a source-shaped term to V_hat_cond; zero design mean does not remove that second-moment imprint.
- NOI-PRED-011: W_hat_cond is available only on the finite strictly positive domain of V_hat_cond; zero never becomes an estimated inverse.
- NOI-PRED-012: S_cond is dimensionless and unavailable wherever the numerator/scale join is incompatible or the scale is not finite and positive.
- NOI-PRED-013: Increasing member count can improve estimation of the stated finite-design target but cannot create exposure or reduce the immutable parent map's realized noise as 1/sqrt(B).
- NOI-PRED-014: A fixed owner-defined linear transformation propagates a conditional second moment as F M F^T; a per-member learned transformation does not obey that fixed-operator identity by implication.
- NOI-PRED-015: A pass for one NOI-owned admission profile neither realizes the next operation nor changes any producer-owned fact.

9. Draft claim boundary

This rationale makes no claim of implementation conformity, representation fidelity, numerical validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization. The selected ordinary route and all numerical products remain unavailable pending their exact parent gates, plan realization, later scientific-owner acceptance, and separately governed downstream evidence.